MILP · Independent exercises

Instructions

Baseline: maximize $8a+6b-10yA-8yB$, subject to $2a+b\le 12$, $a+b\le 10$, $a\le 6yA$, $b\le 8yB$, $a,b\ge 0$ continuous, yA,yB binary. Money: $1,000/week. Reset baseline for each exercise.

For every answer, show the model change, feasible plan, objective, and justification.

1. Integer production

Maximize $3x+2y$ with $x+y\le 4$, $2x+y\le 6.5$, and nonnegative integer x,y. Find the optimum and verify it by enumeration.

Working: _____

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2. Sales cap 6

Reset the plant model to baseline, then replace $a+b\le 10$ with $a+b\le 6$. Compare all four opening patterns.

Working: _____

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3. At most one line

Use baseline data and add $yA+yB\le 1$. What is the best opening and production plan?

Working: _____

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4. A module costing 8

Use baseline plant data plus an optional module adding 4 labor units. Its weekly cost is 8. Compare buy and do not buy.

Working: _____

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5. A minimum batch of 4

Reset to baseline and add a≥4yA. Compare opening patterns and determine the optimum.

Working: _____

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6. A remaining optimality gap

A maximization MILP has feasible incumbent value 46 and a valid global upper bound 48. What can be concluded?

Working: _____

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